

Exploring the Seven Domains of a Typical IT Infrastructure (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 01

Student:

Chidima Onyenobi

Email:

conyenobig@hc.edu

Time on Task:

6 hours, 27 minutes

Progress:

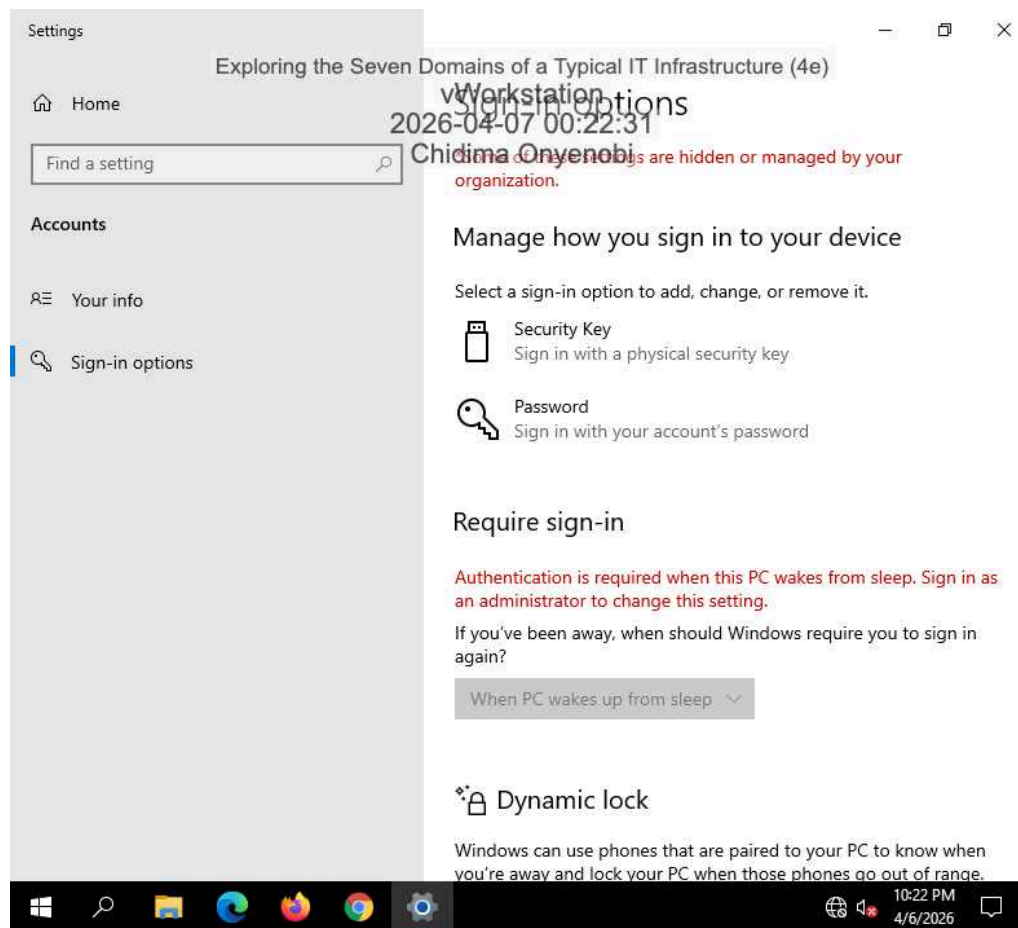
100%

Report Generated: Tuesday, April 7, 2026 at 2:35 PM

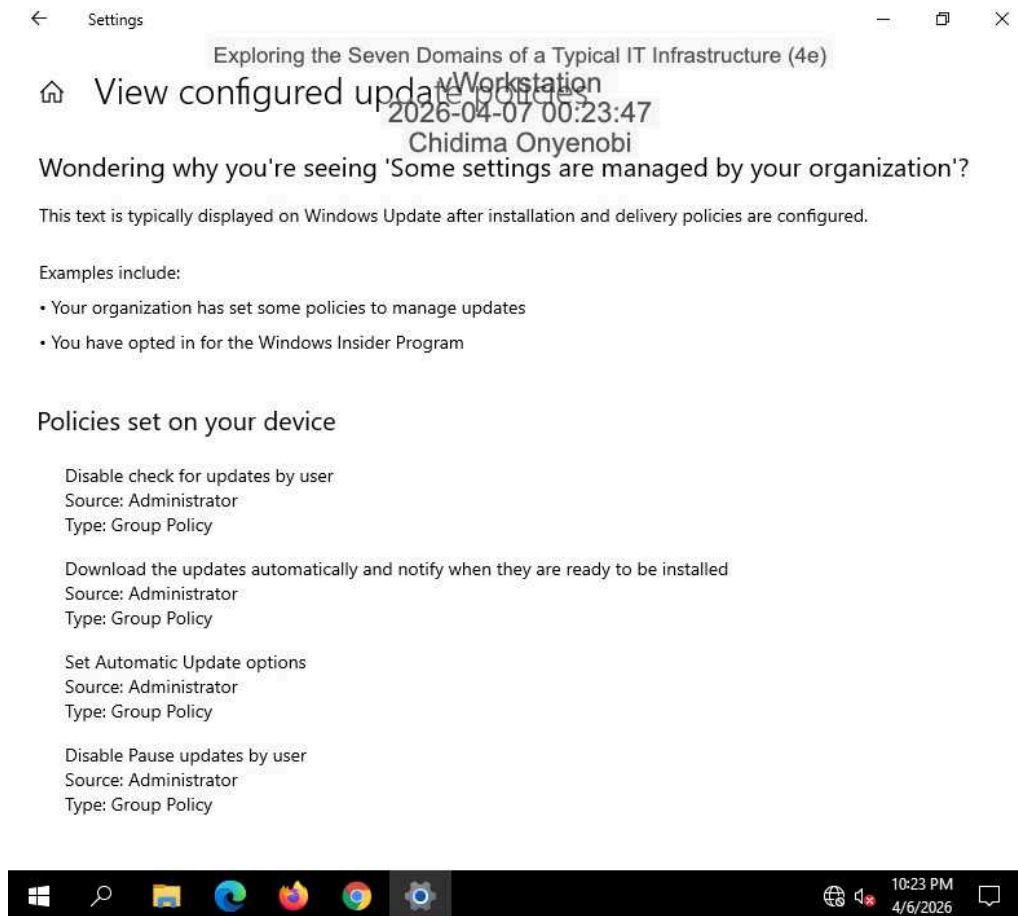
Section 1: Hands-On Demonstration

Part 1: Explore the Workstation Domain

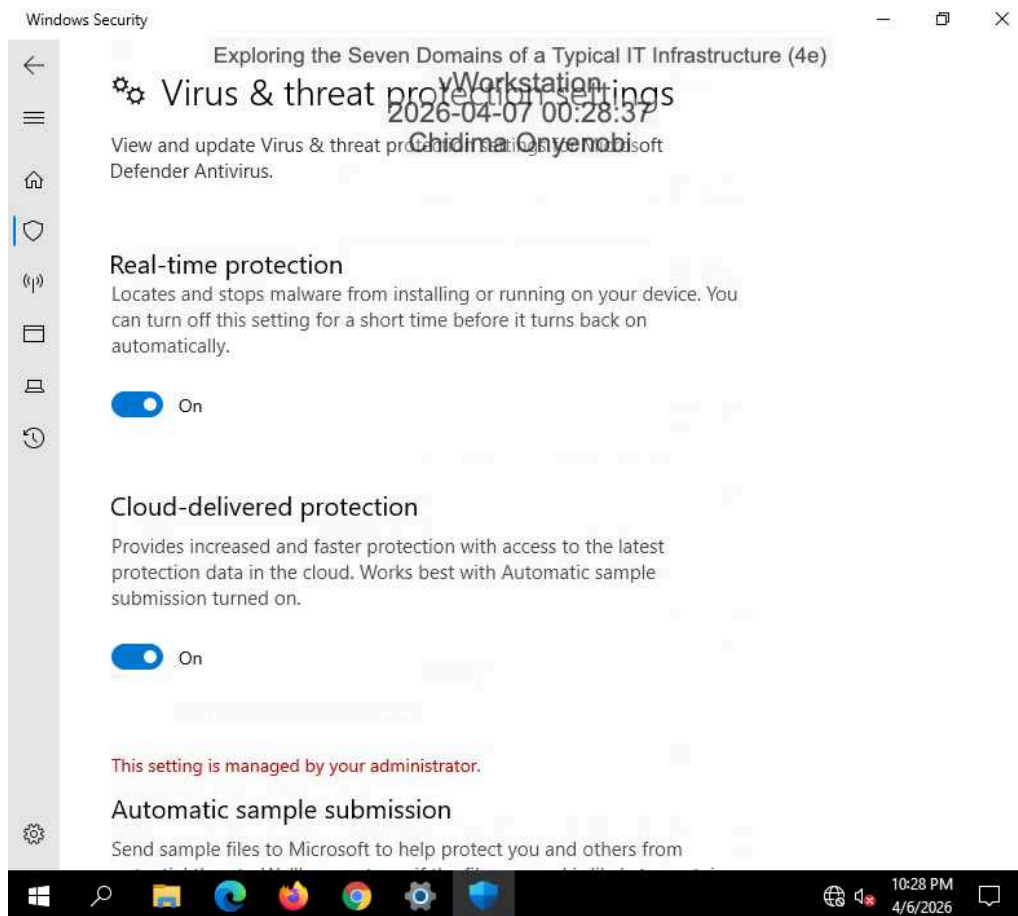
4. Make screen capture showing the **Sign-in options** for Alice's account.



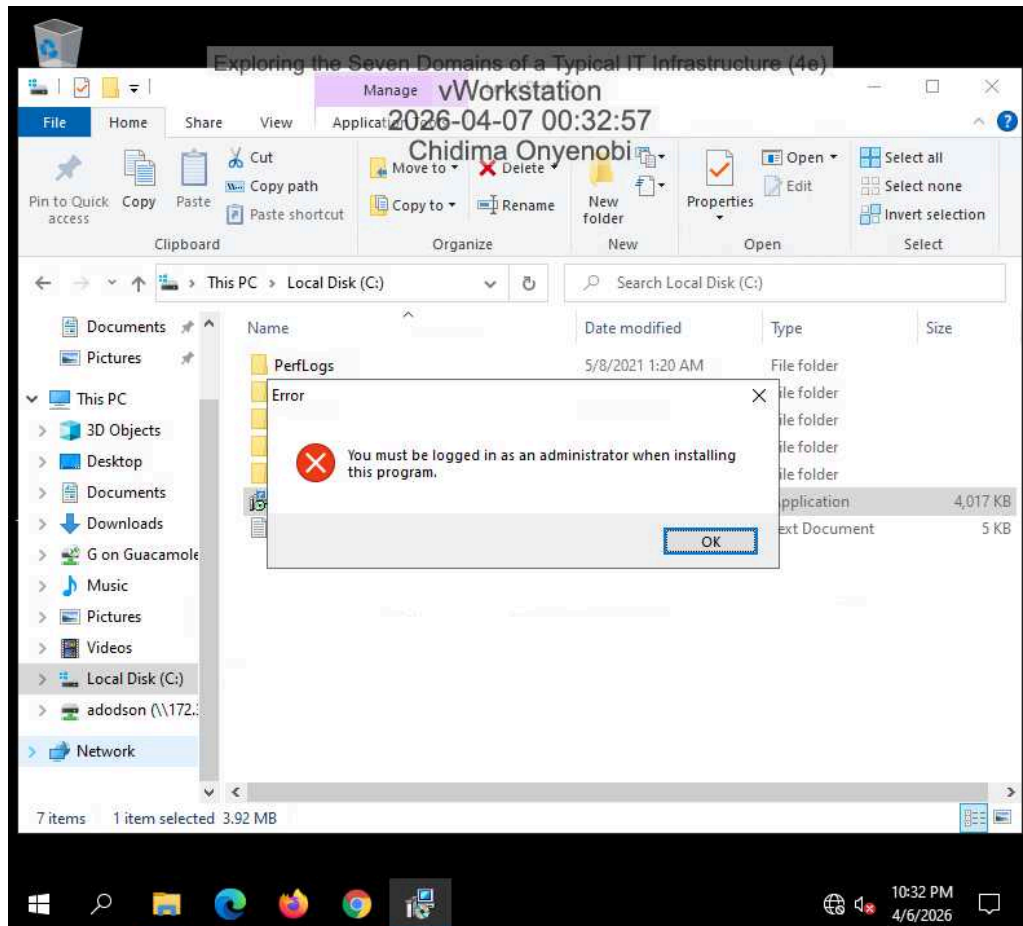
7. Make a screen capture showing the View configured update policies page.



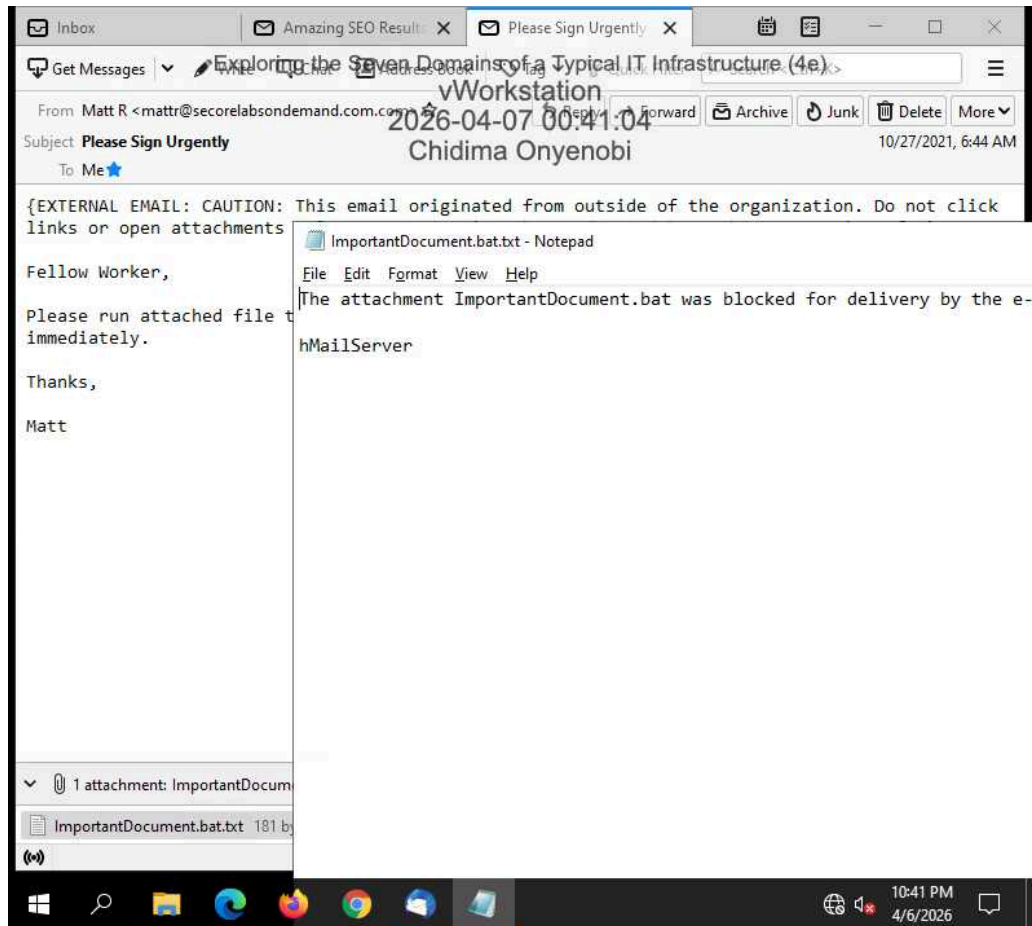
14. Make a screen capture showing the Virus & Threat Protection Settings.



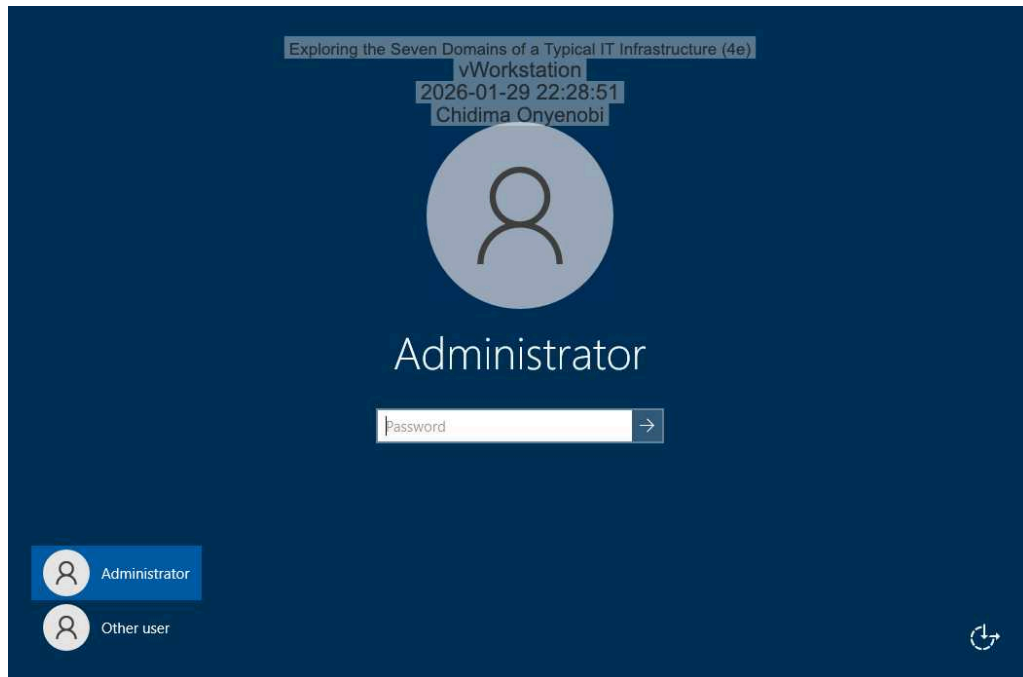
18. Make a screen capture showing the security warning from attempting to run an executable file.



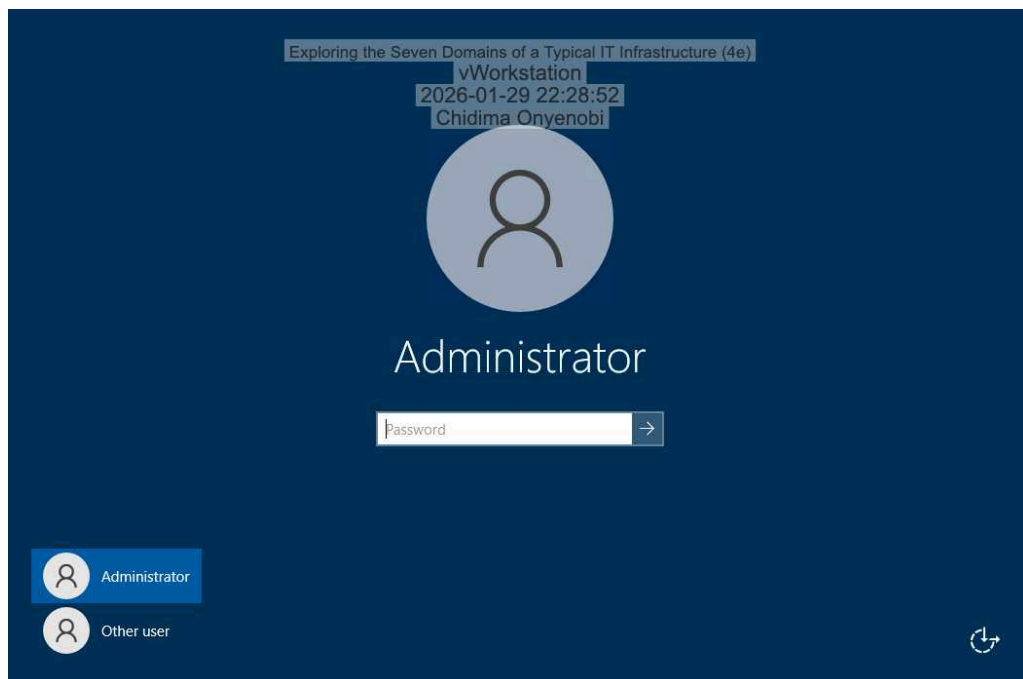
24. Make a screen capture showing the **blocked attachment message**.



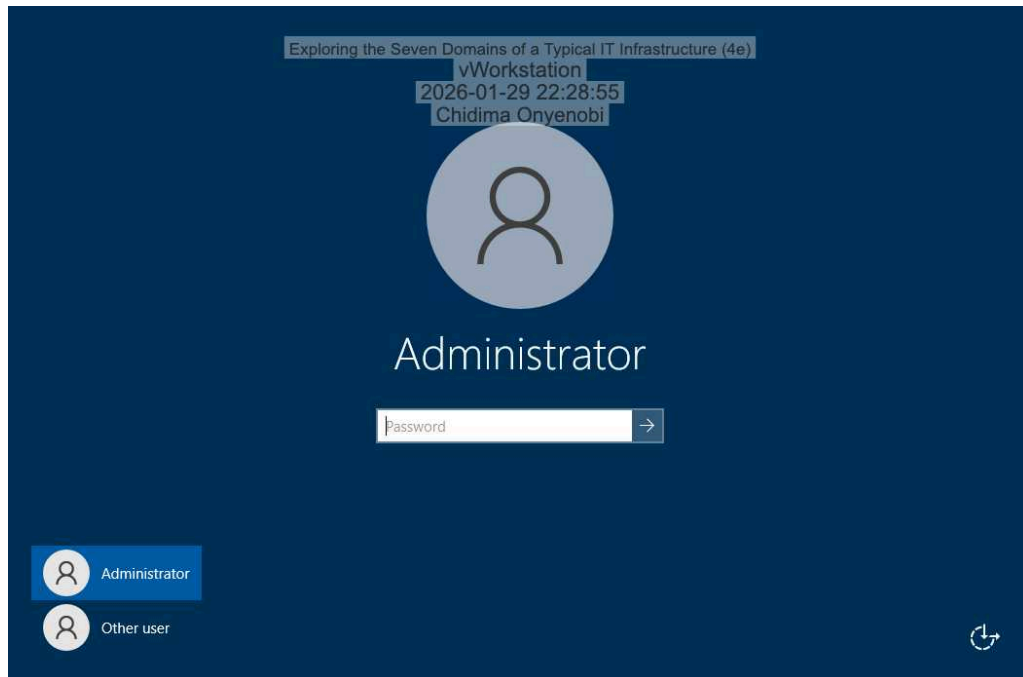
28. Make a screen capture showing a **successful connection to the addodson user folder**.



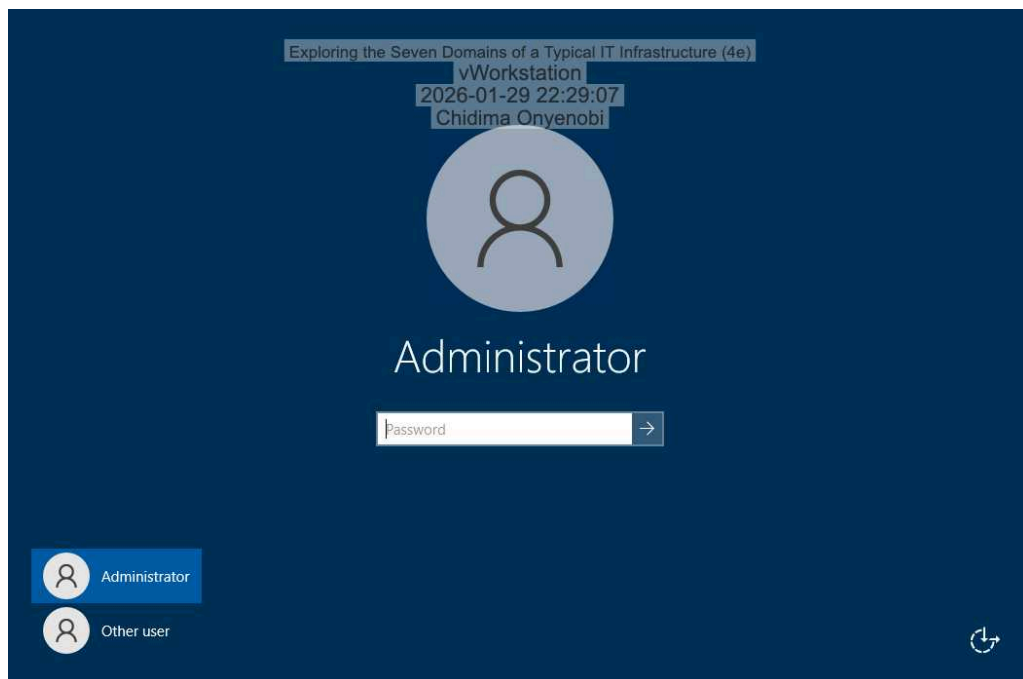
29. Make a screen capture showing a **failed connection to another user folder**.



31. Make a screen capture showing a **successful connection to the Marketing shared folder**.

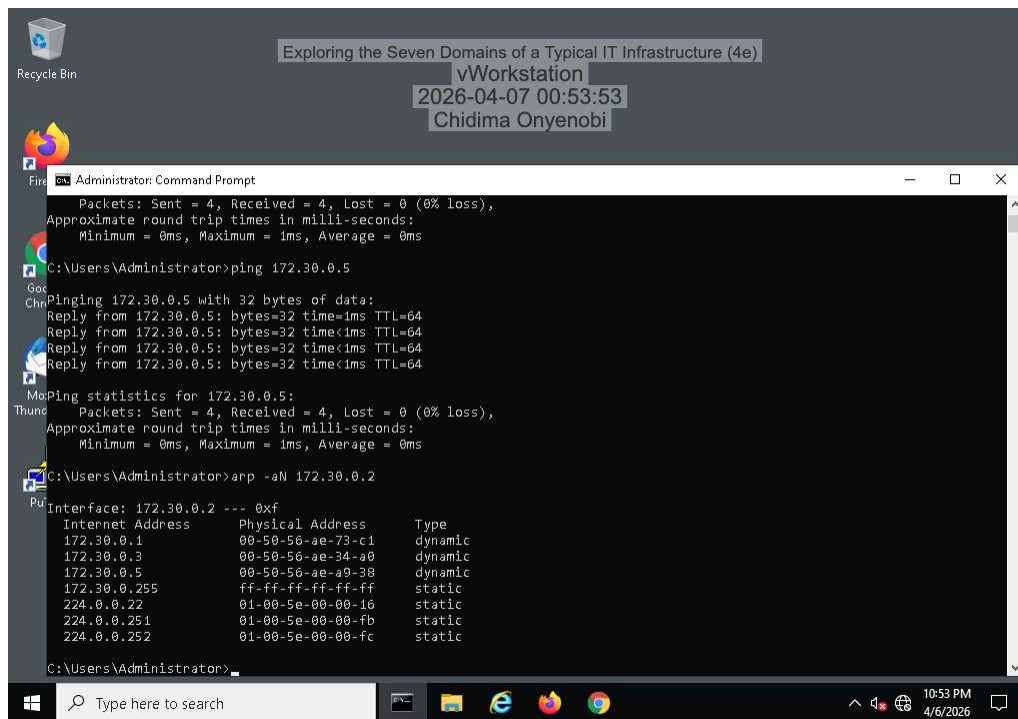


32. Make a screen capture showing a **failed connection to another shared folder**.

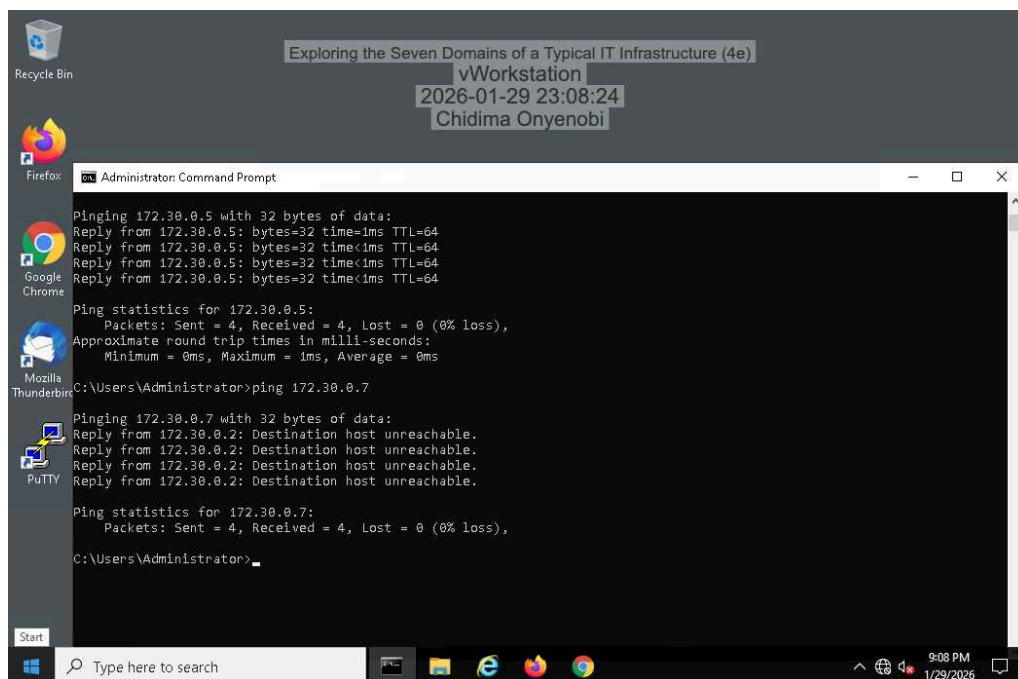


Part 2: Explore the LAN Domain

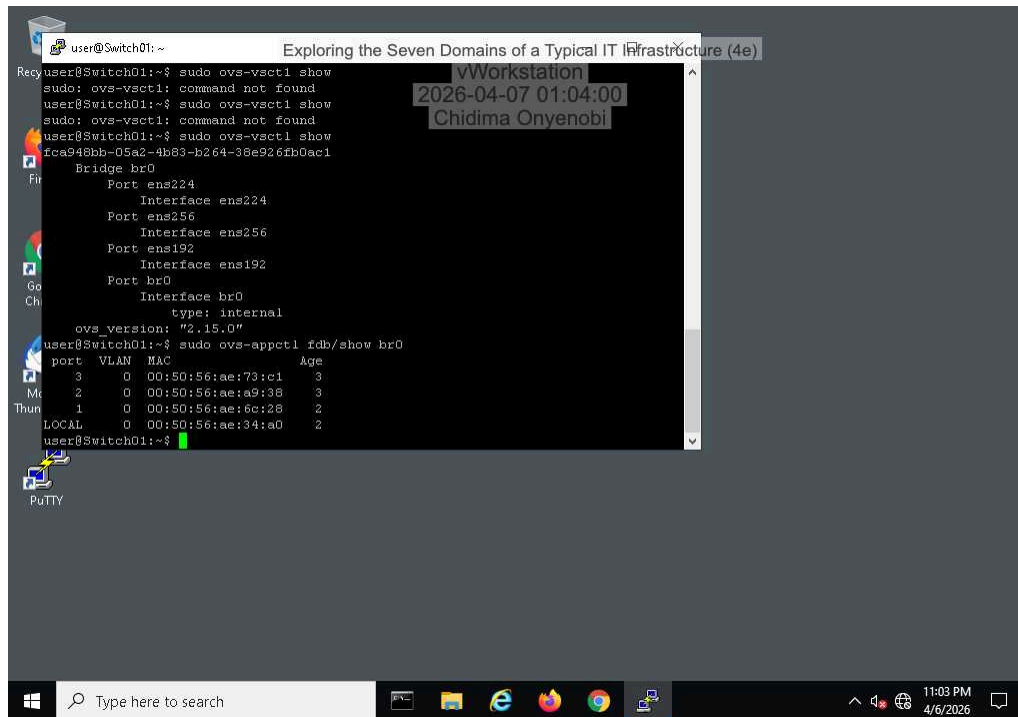
5. Make a screen capture showing the vWorkstation's original ARP table.



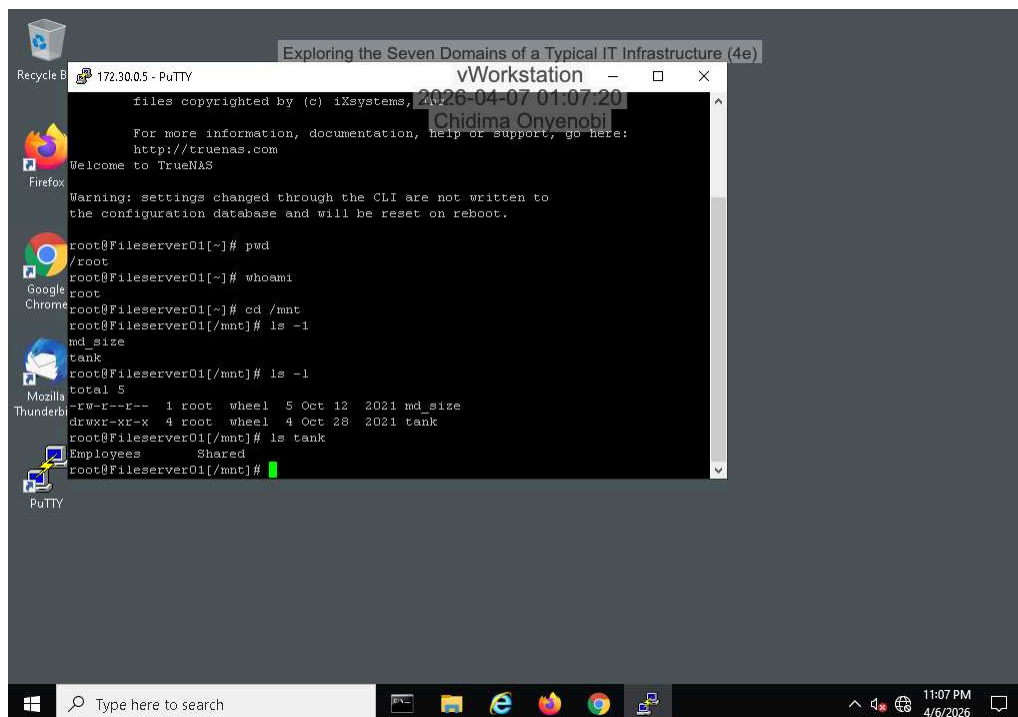
10. Make a screen capture showing the vWorkstation's updated ARP table.



20. Make a screen capture showing the Switch01 forwarding table.

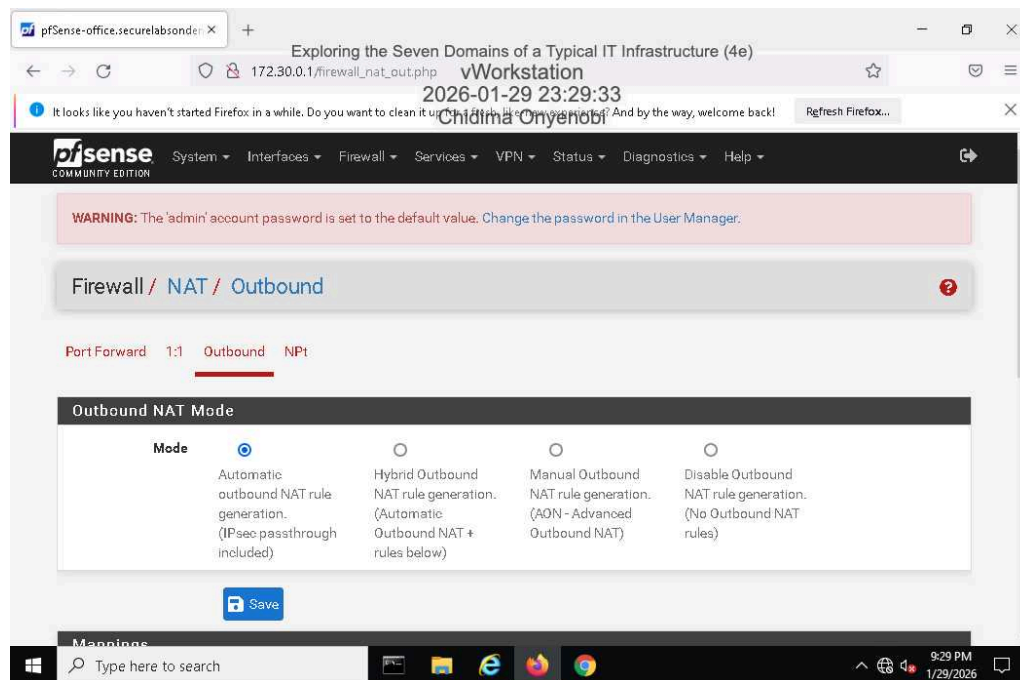


30. Make a screen capture showing the contents of the Employees directory.

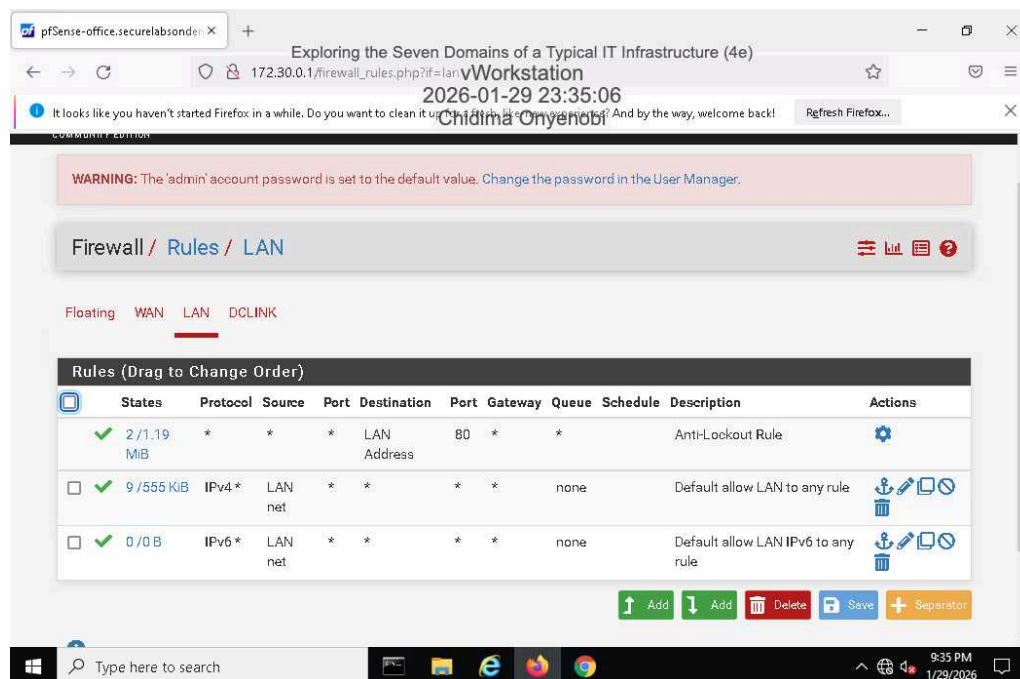


Part 3: Explore the LAN-to-WAN Domain

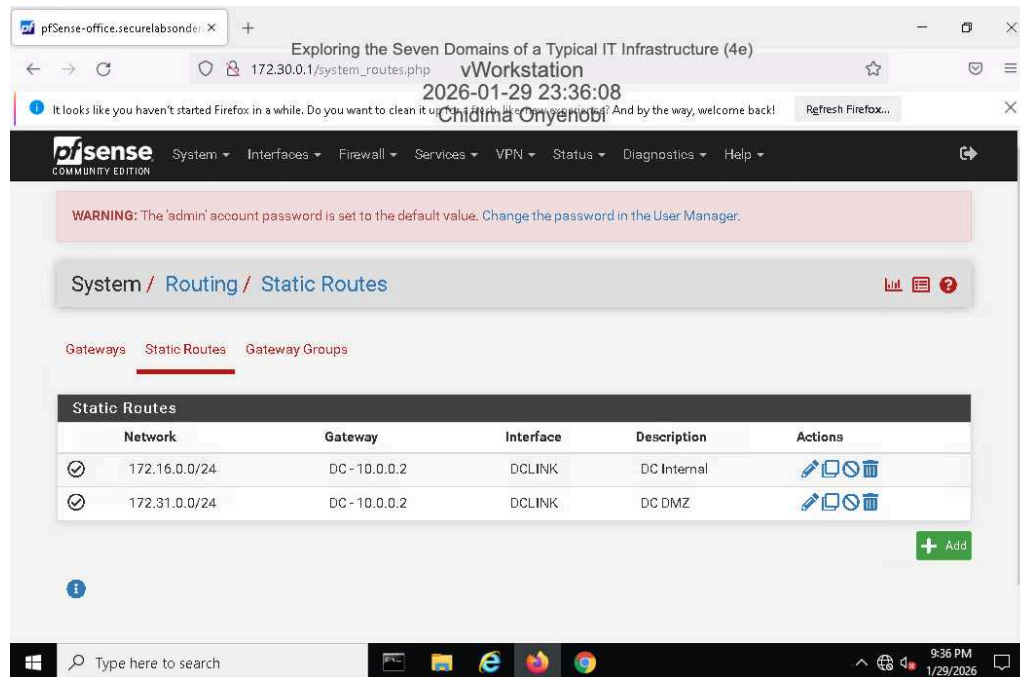
6. Make a screen capture showing the Outbound NAT settings.



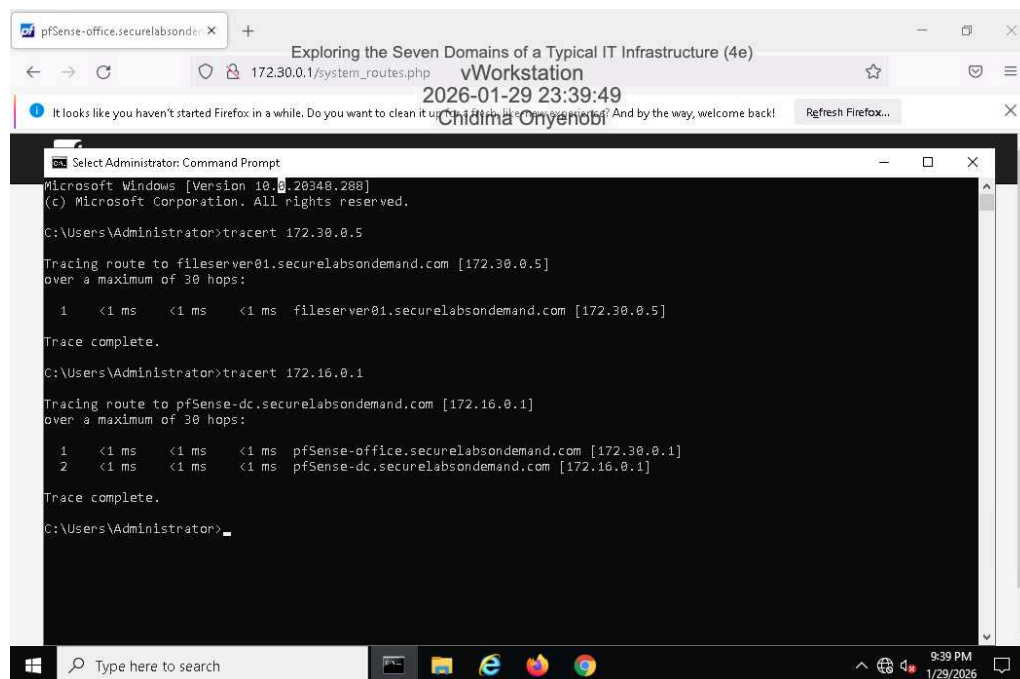
9. Make a screen capture showing the permissive LAN rules.



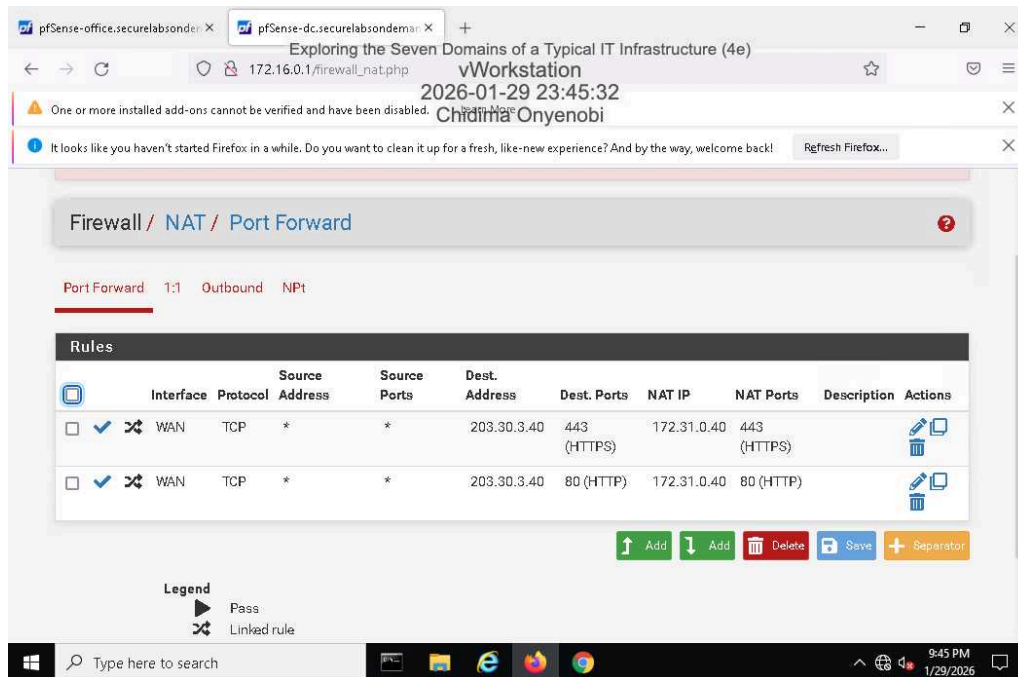
12. Make a screen capture showing the Static Routes page.



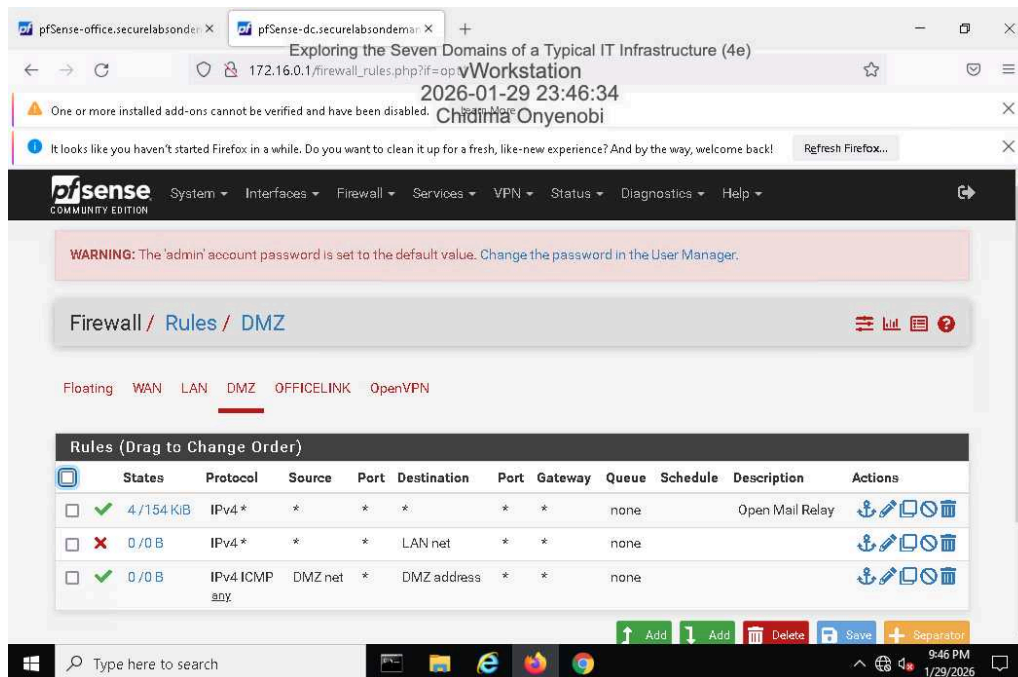
16. Make a screen capture showing the result of your tracert to the pfsense-dc appliance.



22. Make a screen capture showing the Port Forward rules for the web server.



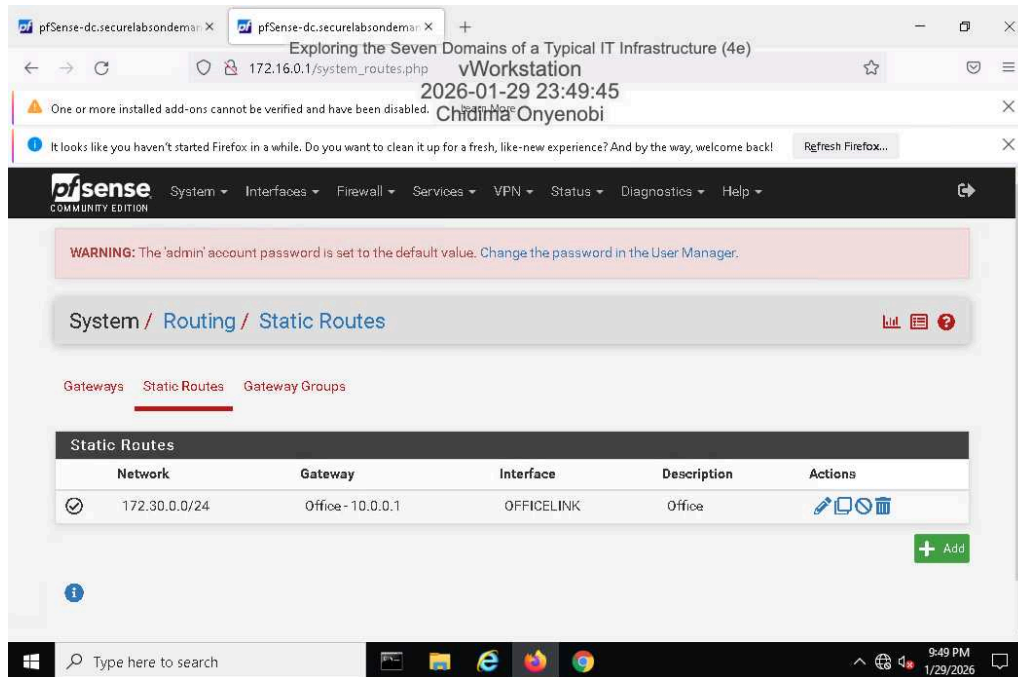
25. Make a screen capture showing the DMZ firewall rules.



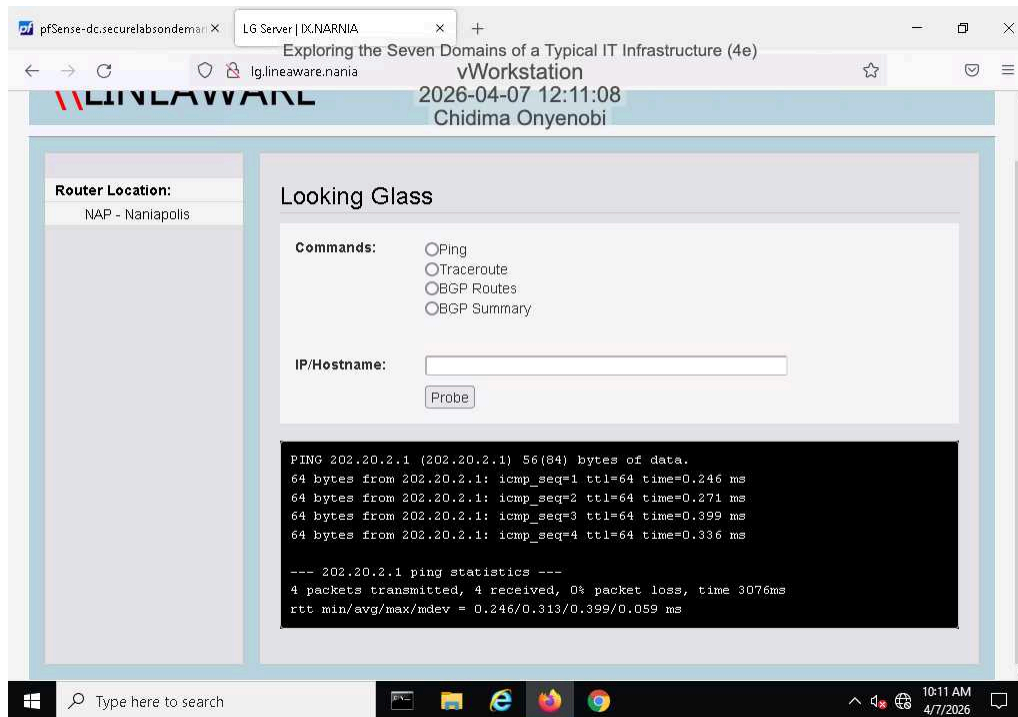
Section 2: Applied Learning

Part 1: Explore the WAN Domain

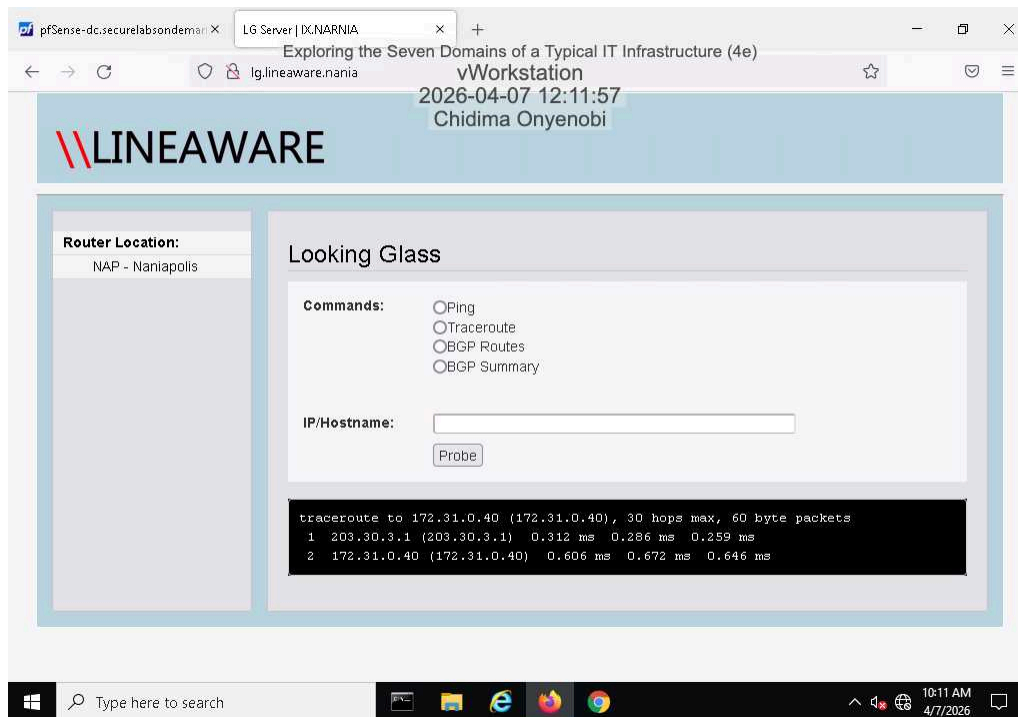
5. Make a screen capture showing the **static route** for the point-to-point connection.



9. Make a screen capture showing the **BPG** neighbor ping results.

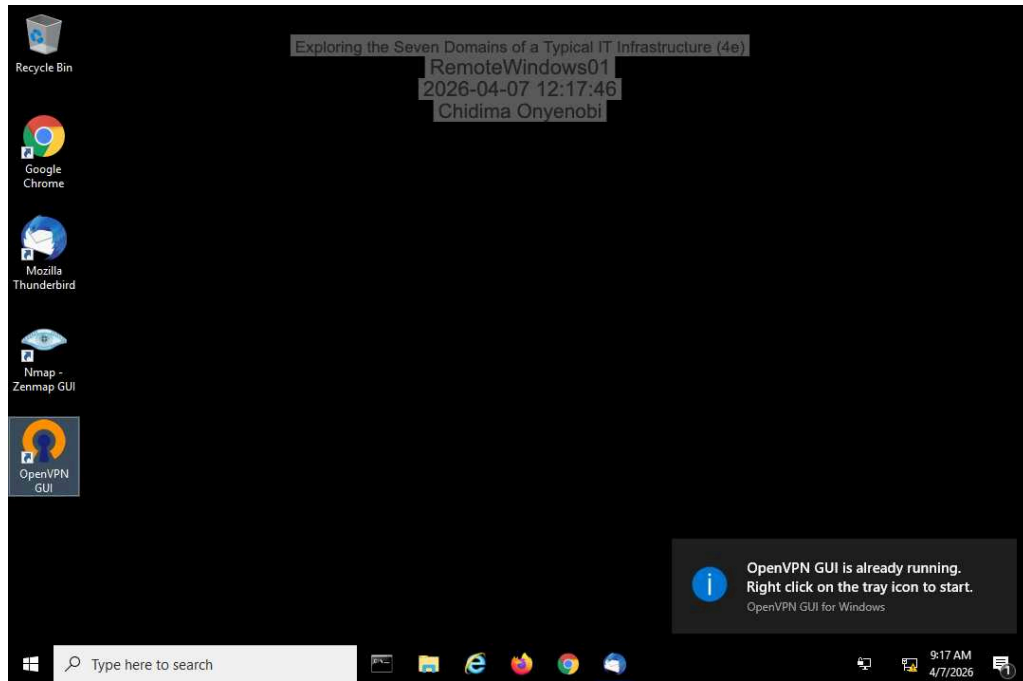


12. Make a screen capture showing the traceroute to the file server.



Part 2: Explore the Remote Access Domain

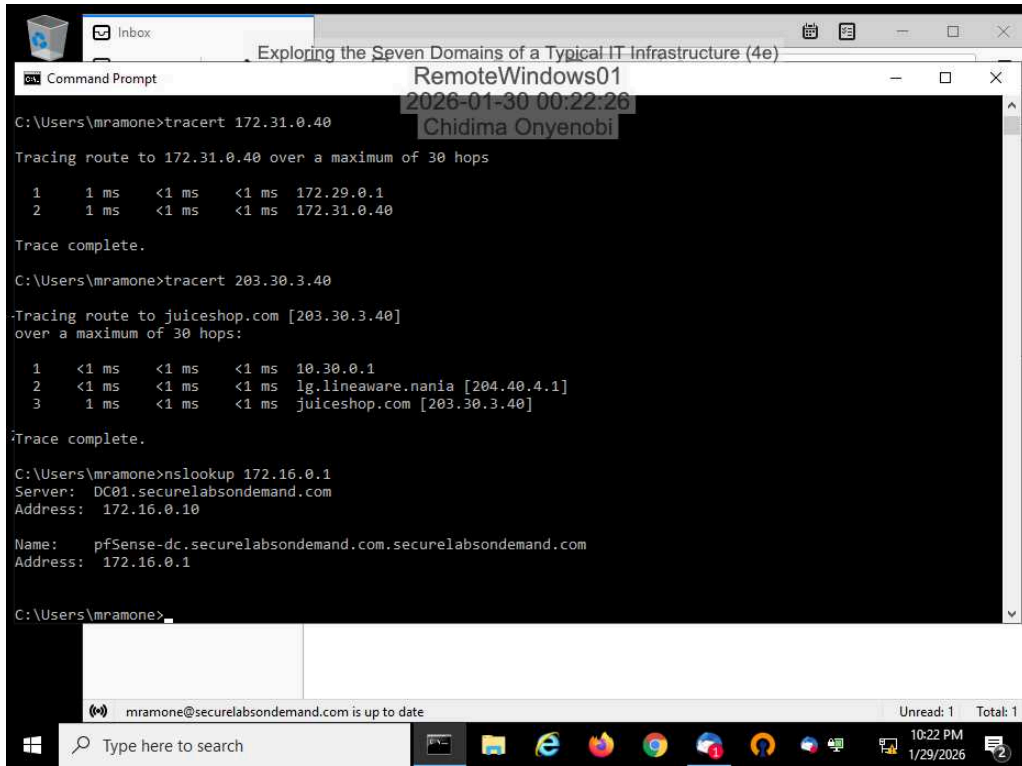
9. **Make a screen capture** showing the **successful connection to the email server**.



14. **Document** whether the VPN connection is split tunnel or full tunnel, based on the tracert results.

The first tracert result was a full tunnel. The second tracert result was a split tunnel.

16. Make a screen capture showing the **successful reverse DNS lookup for the internal host**.



```
C:\Users\mramone>tracert 172.31.0.40

Tracing route to 172.31.0.40 over a maximum of 30 hops:

  1    1 ms    <1 ms    <1 ms    172.29.0.1
  2    1 ms    <1 ms    <1 ms    172.31.0.40

Trace complete.

C:\Users\mramone>tracert 203.30.3.40

Tracing route to juiceshop.com [203.30.3.40]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms    10.30.0.1
  2    <1 ms    <1 ms    <1 ms    lg.lineaware.nania [204.40.4.1]
  3    1 ms    <1 ms    <1 ms    juiceshop.com [203.30.3.40]

Trace complete.

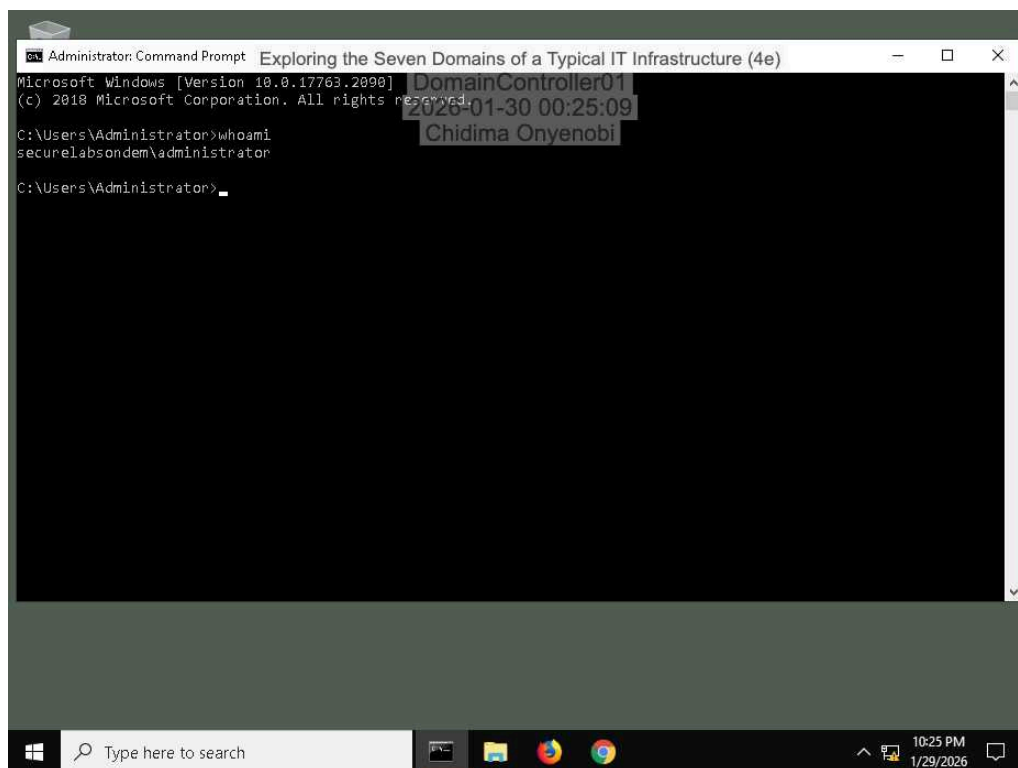
C:\Users\mramone>nslookup 172.16.0.1
Server:  DC01.securelabsondemand.com
Address:  172.16.0.10

Name:    pfSense-dc.securelabsondemand.com.securelabsondemand.com
Address:  172.16.0.1

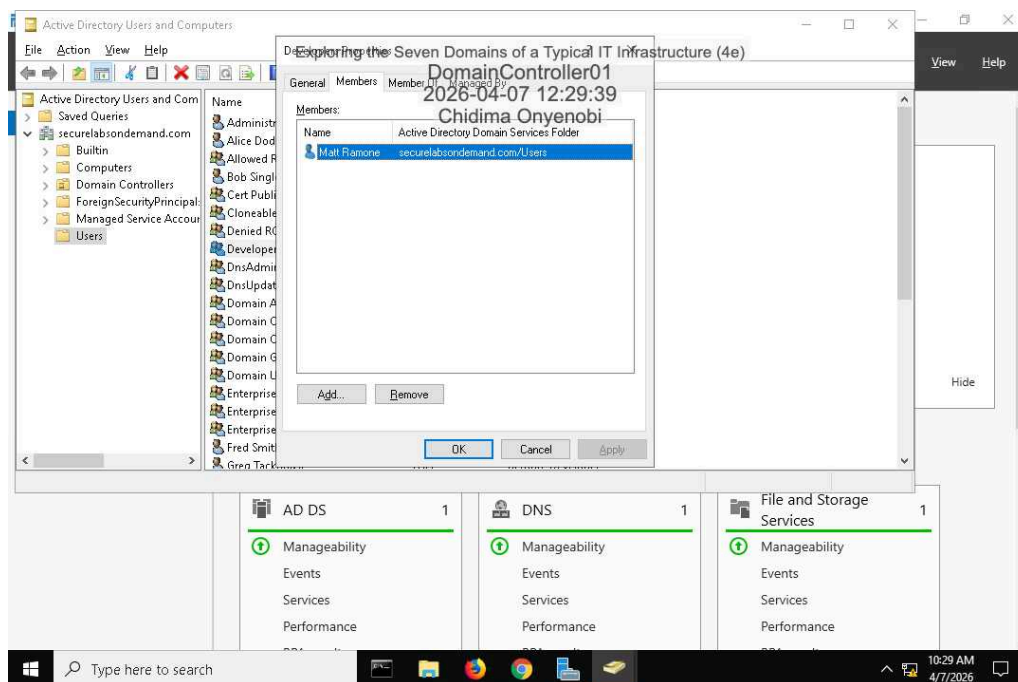
C:\Users\mramone>
```

Part 3: Explore the System/Application Domain

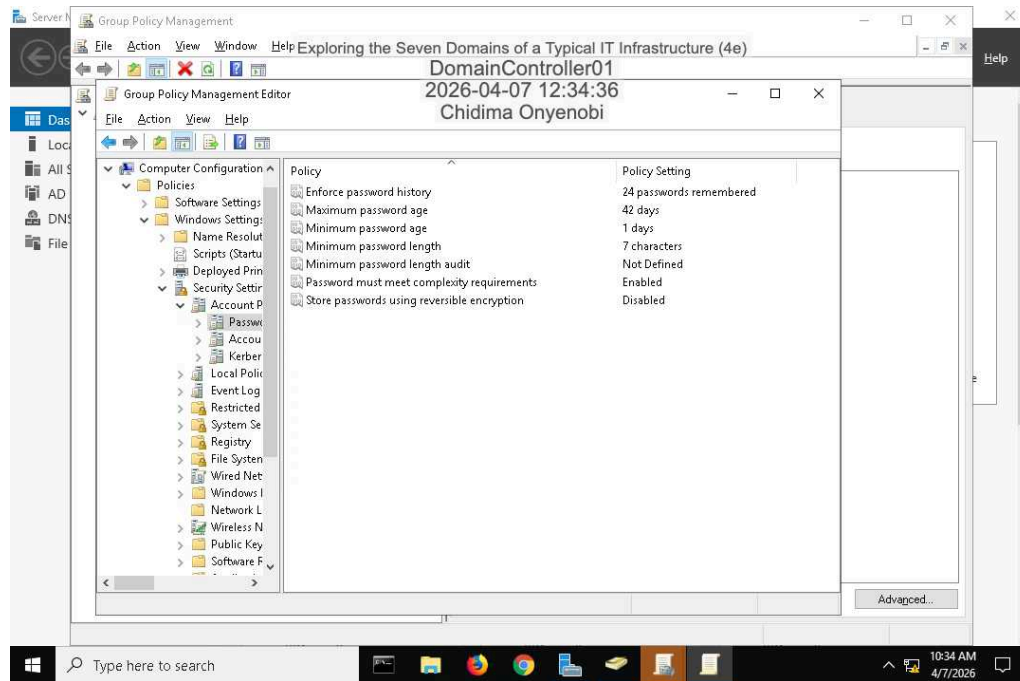
4. Make a screen capture showing the **whoami** results.



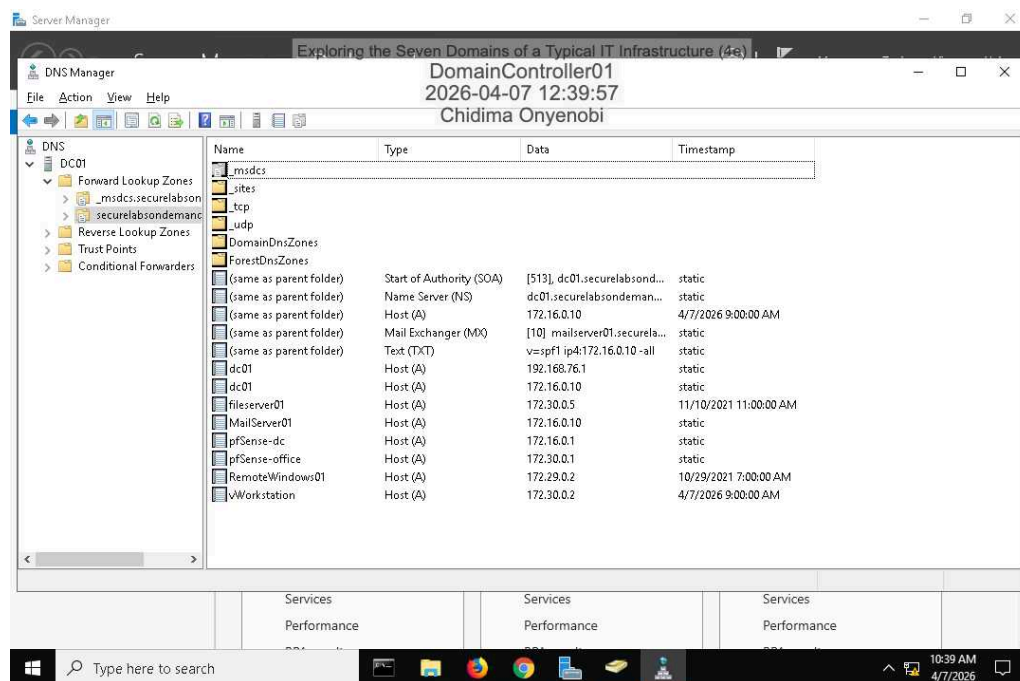
10. Make a screen capture showing the members of the **Developers AD group**.



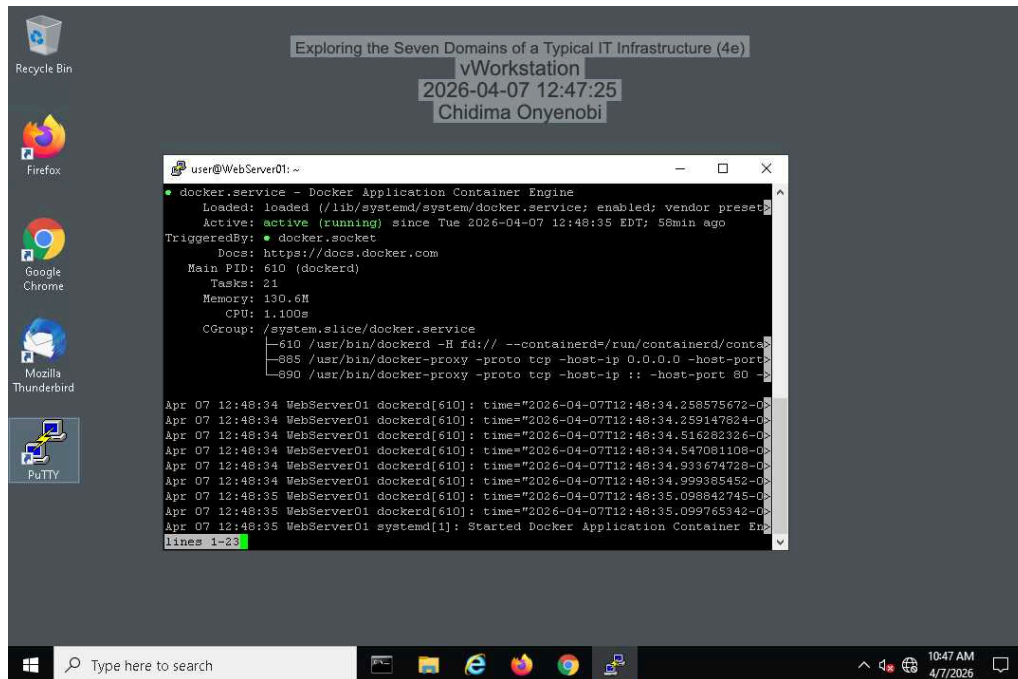
16. Make a screen capture showing the password policy settings in the Group Policy Management Console.



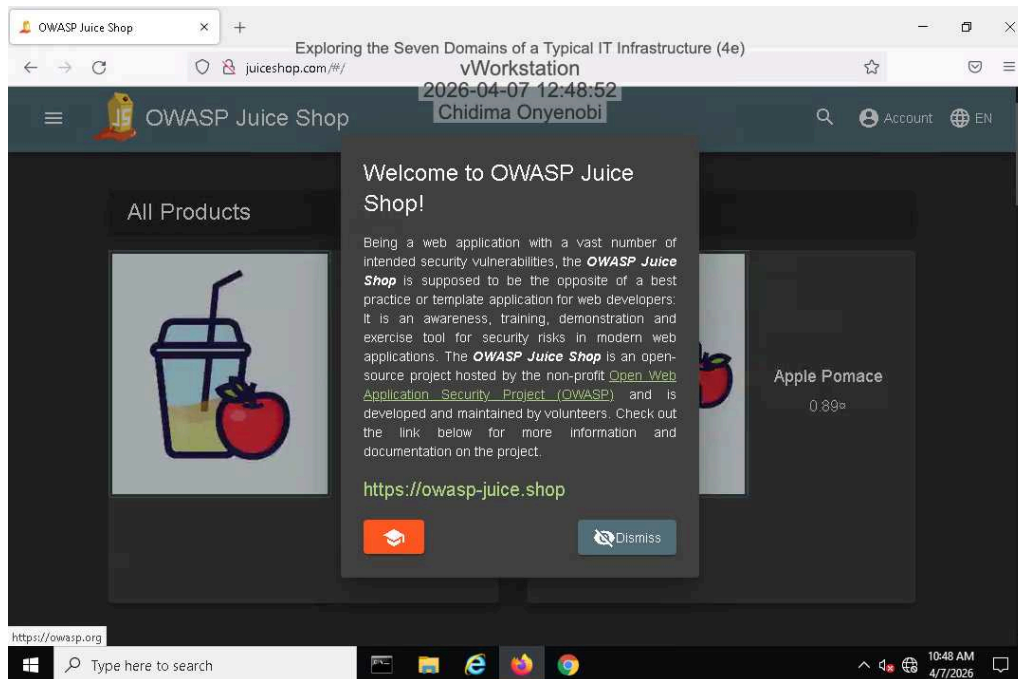
20. Make a screen capture showing the DNS entries.



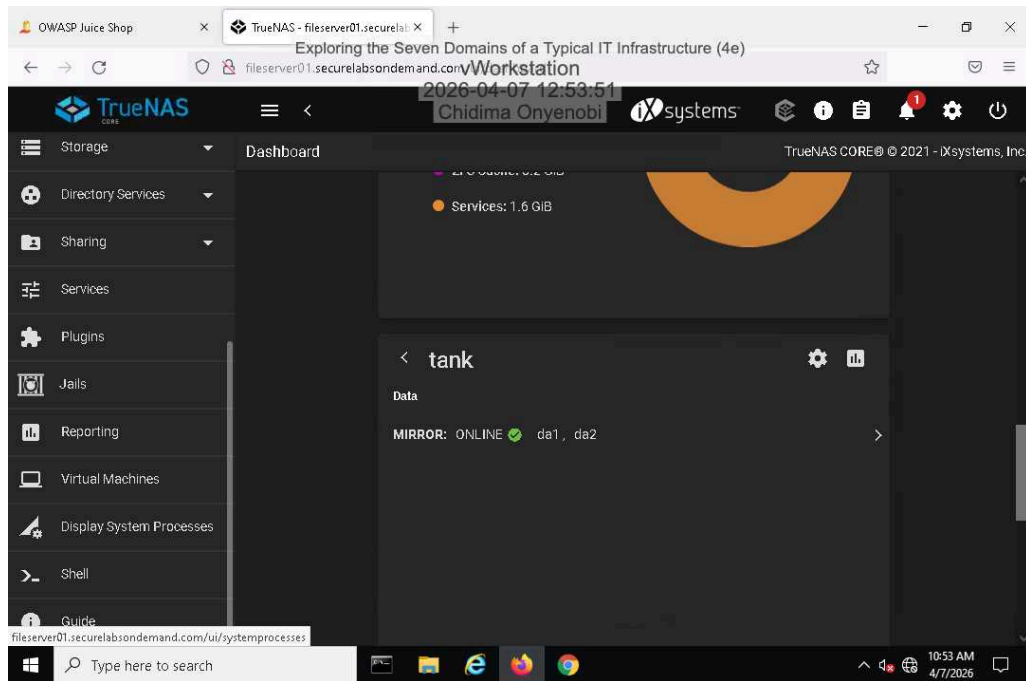
28. Make a screen capture showing the Docker service status.



31. Make a screen capture showing the juiceshop.com web page.



36. Make a screen capture showing the disks in the tank volume.



Section 3: Challenge and Analysis

Part 1: Explore the User Domain

Based on your research, **identify** at least **two compelling threats** to the User Domain and **two effective security controls** used to protect it. Be sure to cite your sources.

-Two compelling threats are phishing and insider threats. In phishing, attackers can use emails to trick you into revealing your personal information. An insider threat would be when an employee misuses their authorized access.-Security controls should include security awareness training to educate users on recognizing phishing attempts and using multi-factor authentication to require 2 or more forms of verification; that way, if the account gets hacked, the hacker cannot access personal or confidential information.

Part 2: Research Additional Security Controls

Based on your research, **identify** security controls that could be implemented in the Workstation, LAN, LAN-to-WAN, WAN, Remote Access, and System/Application Domains. **Recommend** and **explain** one security control for each domain. Be sure to cite your sources.

-A workstation-endpoint protection platform would be needed because it provides tools including antivirus, anti-malware, and host-based firewalls.-LAN network segmentation would be recommended. By dividing the network into smaller sections, you limit the radius of potential breaches in the system.-A LAN-to-WAN IPS would be recommended because it automatically blocks suspicious traffic, preventing threats.-WAN-site-to-site VPN encryption. Using a VPN would ensure that data moving between different company spots is encrypted.Remote Access—multi-factor authentication would be recommended. This defense is the most effective against attackers who seek to compromise credentials.-System/Application Domains- An automated patch management would be perfect because it ensures that the operating system and software are updated to address unknown problems.